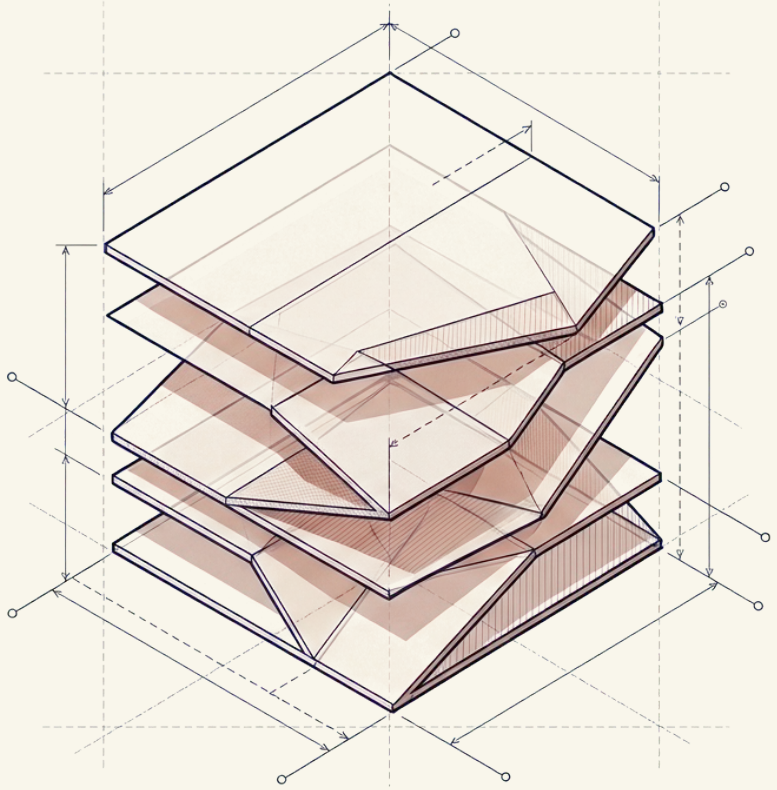
SYSTEM-LEVEL DRIFT MEASUREMENT

The Characteristic Drift Spectrum

A task-free, multiscale fingerprint of systemic organization in complex systems

Systemic drift, not just more data from the same system.

Structural drift | Information drift | Kolmogorov drift



Ishanu Chattopadhyay

University of Kentucky

THE PROBLEM

More data is not drift

Two samples from an unchanged system never match row for row. A drift measure must ignore this re-realization noise and respond when the system's organization changes.

Illustrative example: two survey items A and B, n = 1,000 per sample

SYSTEM P · SAMPLE 1

same organization

	B yes	B no
A yes	396	104
A no	97	403

Yes-rates of A, B: **0.500, 0.493**

P(B = yes | A = yes / no)

0.792 / 0.194

SYSTEM P · SAMPLE 2

same organization, new draw

	B yes	B no
A yes	412	93
A no	104	391

Yes-rates of A, B: **0.505, 0.516**

P(B = yes | A = yes / no)

0.816 / 0.210

SYSTEM Q · SAMPLE

items decouple; yes-rates unchanged

	B yes	B no
A yes	247	254
A no	251	248

Yes-rates of A, B: **0.501, 0.498**

P(B = yes | A = yes / no)

0.493 / 0.503

Marginal checks: the resample shifts more (up to 0.023) than the reorganized system (at most 0.005).

Conditional relation: 0.79 → 0.82 on resampling, but 0.79 → 0.49 when the items decouple.

Drift should mean the system changed, not that we drew new data.

Re-realization or reorganization?

We want a drift measure that is a property of the **system**, not of the particular sample: its estimate should settle as data accumulate, and move only when the organization moves.

SAME ORGANIZATION P

Re-realization

Independent samples differ only as realizations. With stable, consistent conditional learning and estimation:

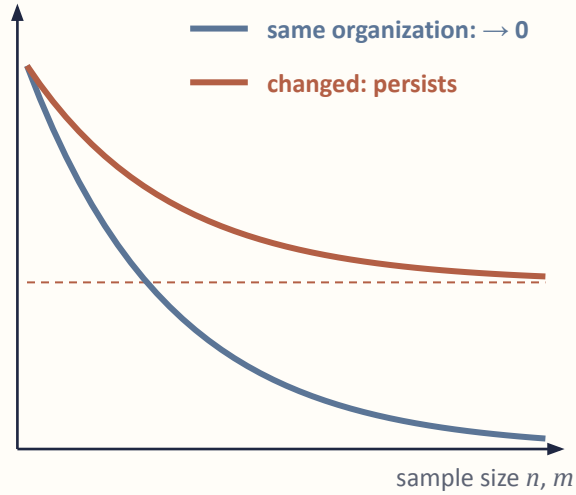
$$d\big(\widehat{\mathcal{S}}_n(P),\, \widehat{\mathcal{S}}_m(P)\big) \;\longrightarrow\; 0, \qquad n, m \to \infty$$

CHANGED ORGANIZATION $P \to Q$

Reorganization

If the change alters the conditional organization, the residual hierarchy, or the information structure the probes interrogate, then $S(P)$ and $S(Q)$ need not coincide. The displacement persists rather than vanishing as a finite-sample effect.

Distance between estimated spectra



schematic

Repeated realizations of the same organization approach the same fingerprint; organizational drift deforms that fingerprint.

Existing checks answer a different question

Approach	New sample, same system	Reorganized system	What it tells you
Per-feature marginal checks	Sampling fluctuations trigger alarms across many features unless corrected	Can miss reorganization that leaves marginals intact	Which variables' levels moved
Model-performance (concept) drift [1, 2]	Stable up to sampling noise	Seen only if it affects the chosen response Y	One task's performance; needs labels
Global two-sample and classifier tests [3, 4]	False alarms held at the test's level	Detects that the distributions differ	One score: not what changed, or where
Characteristic drift spectrum	**Converges to the same fingerprint as data accumulate**	**Persistent spectral displacement**	**Which probe moved, at which depth; no Y needed**

Scope. The spectrum is not an injective encoding of the joint distribution: distinct distributions can agree on a finite set of probes. Its purpose is operational: to separate persistent change in the probed organization from finite-sample re-realization, and to record where in the hierarchy it appears.

Learning the conditional organization

For categorical state $X = (X_1, \ldots, X_p)$ and row population R, a conditional learner returns target-wise predictors φ_i of X_i given X_{-i}:

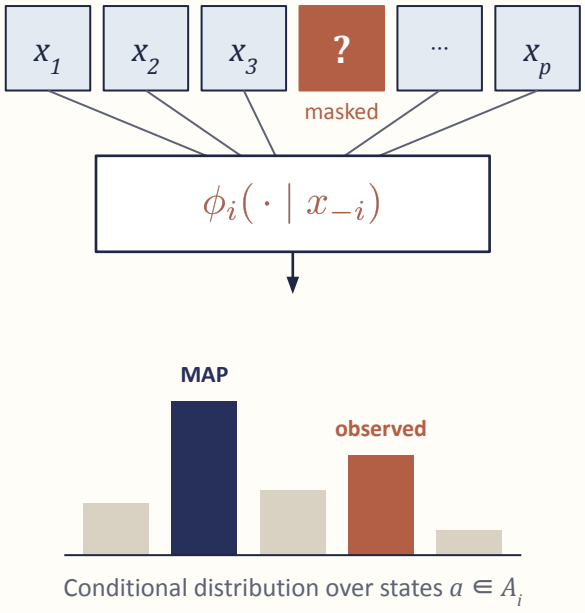
$$M(R) = \{\phi_i(\cdot \mid x_{-i})\}_{i \in T_M}$$

For row x, the mean conditional log-persistence is

$$P_M(x) = \frac{1}{n_M(x)} \sum_{i \in \mathcal{I}_M(x)} \log_2 \left[\max\{\phi_i(x_i \mid x_{-i}), \varepsilon\} \right]$$

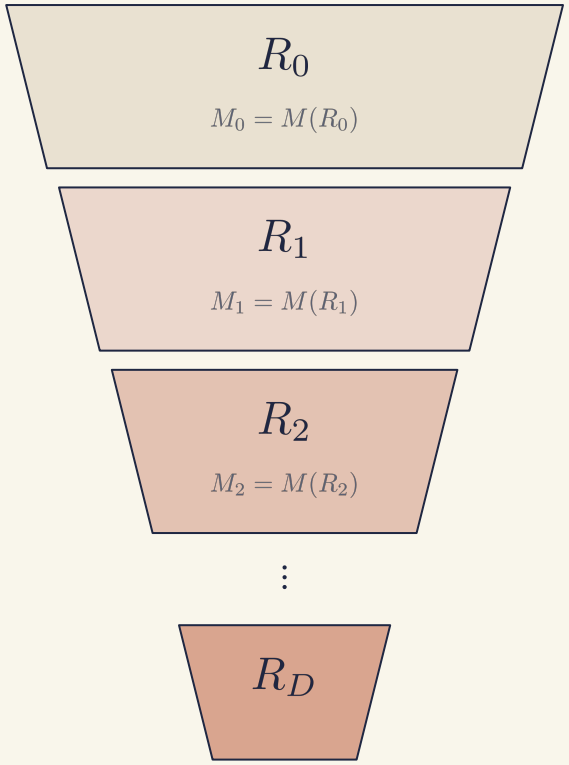
$P_M(x)$ is the average log support assigned to the observed coordinates when each coordinate is predicted from the remainder of the state.

Low persistence identifies rows that are poorly reconstructed by the dominant learned conditional organization.



Learner-agnostic. The objective is the full family of conditionals, not one selected response. Any sufficiently accurate and stable conditional learner can instantiate the spectrum; the GSS results use Large Science Models (LSMs) [5, 6].

Peeling recovers a residual hierarchy



$$R_0 \supset R_1 \supset \cdots \supset R_D$$

a freshly fitted conditional model at every level

1 **Fit and score**

At depth d, fit a conditional model $M_d = M(R_d)$ and compute the persistence of every row under M_d.

2 **Robustly standardize lack of persistence**

$$Z_d^{\text{lack}}(x) = \frac{c_d - P_{M_d}(x)}{s_d}$$

$$c_d = \operatorname{median} P_{M_d}, \qquad s_d = \kappa_{\text{MAD}} \operatorname{median} |P_{M_d} - c_d|$$

$$\kappa_{\text{MAD}} = \{\Phi^{-1}(3/4)\}^{-1} \approx 1.4826$$

3 **Peel and refit**

$$R_{d+1} = \{x \in R_d : Z_d^{\text{lack}}(x) > z_*\}, \qquad z_* = 0.5$$

Intuition. Peeling removes the statistically dominant core and refits on the surviving low-persistence population, so organization masked at the population level becomes visible. Depth is structural, not chronological.

Three nonredundant probes at every depth

PROBE I

Δ_C

Structural drift

ASKS

Does MAP-normalized reconstruction change after refitting?

SENSITIVE TO

Reconstruction geometry

PROBE II

Δ_K^{pseudo}

Information drift

ASKS

Does the residual population have a different conditional description length under the parent organization?

SENSITIVE TO

Conditional description length

PROBE III

Δ_K^{gzip}

Kolmogorov drift

ASKS

Are adjacent residual populations distinguishable through compression, without the fitted model as the measuring instrument?

SENSITIVE TO

Model-independent compressibility

Why three? The probes need not move together: information drift can be nonzero even when structural drift is zero. Collapsing them to one coordinate would discard distinguishable forms of reorganization.

Structural drift

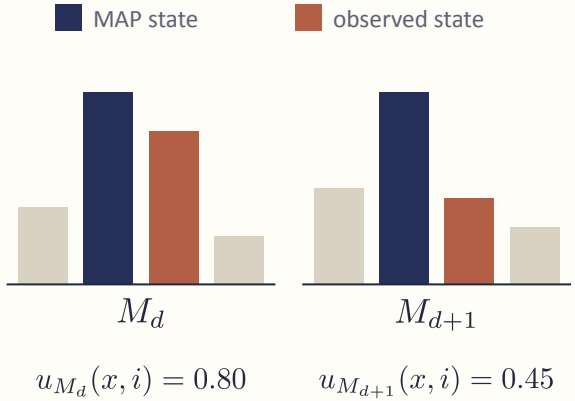
Δ_C

MAP-normalized masked-state reconstruction score

$$u_M(x,i) = \frac{\phi_i(x_i \mid x_{-i})}{\max_{a \in \mathcal{A}_i} \phi_i(a \mid x_{-i})}$$

u_M = 1 when the observed state attains the conditional MAP support; smaller values measure its support relative to that MAP state.

With a balanced probe cloud Q_d and common evaluable targets $I_d(x)$, average the change between adjacent depths:



$$\Delta_C^{(d)} = \frac{1}{|Q_d|} \sum_{x \in Q_d} \frac{1}{|\mathcal{I}_d(x)|} \sum_{i \in \mathcal{I}_d(x)} \left| u_{M_d}(x,i) - u_{M_{d+1}}(x,i) \right|$$

Intuition. Structural drift measures how the system's learned ability to reconstruct masked coordinates changes across structural depth.

Information drift

Δ_K^{pseudo}

A data-adaptive code from the learned conditional structure

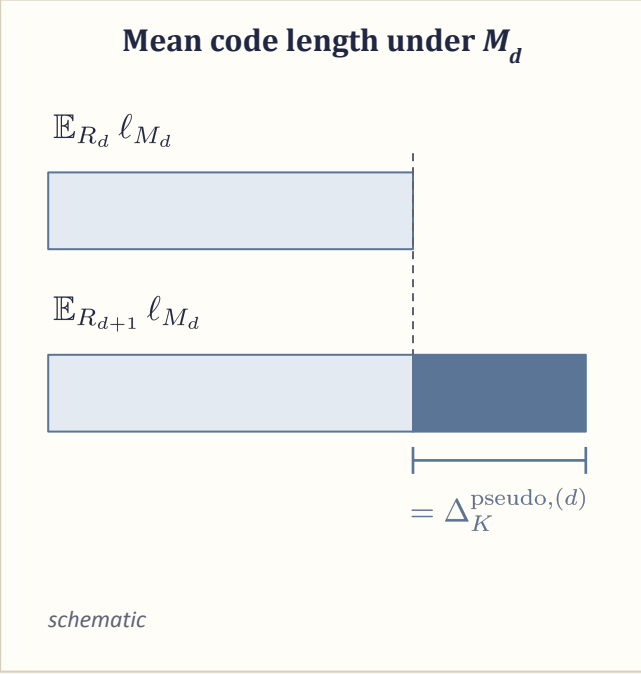
$$\ell_M(x) = -\frac{1}{n_M(x)} \sum_{i \in \mathcal{I}_M(x)} \log_2 \big[\max\{\phi_i(x_i \mid x_{-i}), \varepsilon\} \big]$$

Mean code length (bits) of row x when each coordinate is encoded with the fitted conditional predictor.

Information drift between adjacent residual levels

$$\Delta_K^{\text{pseudo},(d)} = \mathbb{E}_{R_{d+1}} \ell_{M_d} - \mathbb{E}_{R_d} \ell_{M_d}$$

Both expectations use the parent model M_d: rows of R_{d+1} are coded under the organization learned on R_d.



Intuition. A conditional-model-based description-length probe: information displacement is measured using structure learned from the system itself.

PROBE III

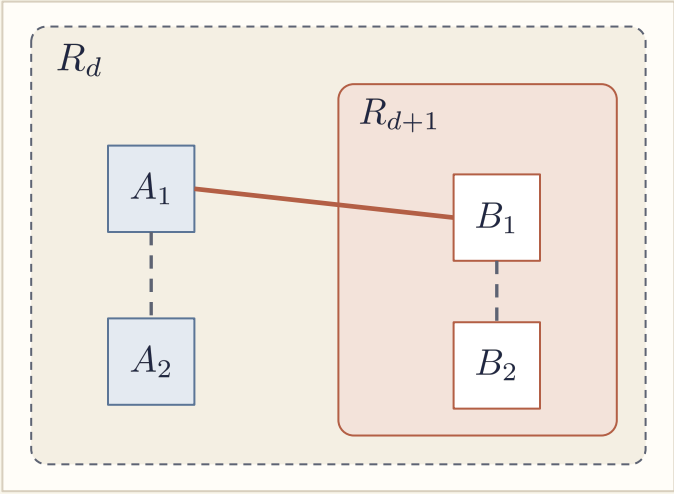
Kolmogorov drift

Δ_K^{gzip}

At depth d, draw matched row blocks $A_1, A_2 \subset R_d$ and $B_1, B_2 \subset R_{d+1}$. For serialized blocks a, b and compressor length $C(\cdot)$, the normalized compression distance is

$$\mathrm{CD}(a,b) = \frac{C(a \,\|\, b) - \min\{C(a), C(b)\}}{\max\{C(a), C(b)\}}$$

Within-depth distances establish the intrinsic compression scale; the across-depth distance measures separation between adjacent levels:



$$\Delta_K^{\mathrm{gzip},(d)} = \mathbb{E}\Big[\mathrm{CD}(A_1, B_1) - \tfrac{1}{2}\big\{\mathrm{CD}(A_1, A_2) + \mathrm{CD}(B_1, B_2)\big\}\Big]$$

Intuition. Compression-calibrated information displacement between adjacent residual levels, without using the fitted conditional model as the measuring instrument.

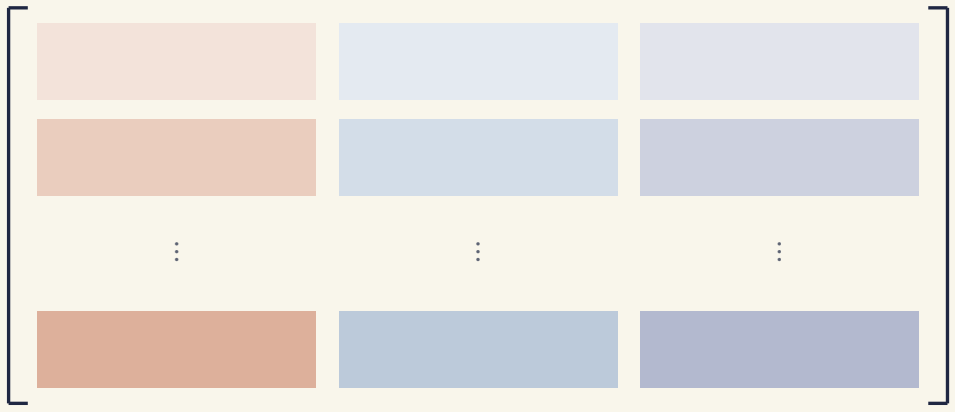
SYNTHESIS

The characteristic drift spectrum

At every depth d, the three numbers form $\mathcal{D}_X(d) = \left(\Delta_C^{(d)},\ \Delta_K^{\mathrm{pseudo},(d)},\ \Delta_K^{\mathrm{gzip},(d)}\right)$

and stacking them across depth gives a depth-by-coordinate matrix:

$$M_X = \begin{bmatrix} \Delta_C^{(d)} & \Delta_K^{\mathrm{pseudo},(d)} & \Delta_K^{\mathrm{gzip},(d)} \\ \hline \\ \\ \vdots & \vdots & \vdots \\ \\ \\ \hline \end{bmatrix} \begin{matrix} \mathcal{D}_X(0) \\ \mathcal{D}_X(1) \\ \vdots \\ \mathcal{D}_X(D-1) \end{matrix}$$

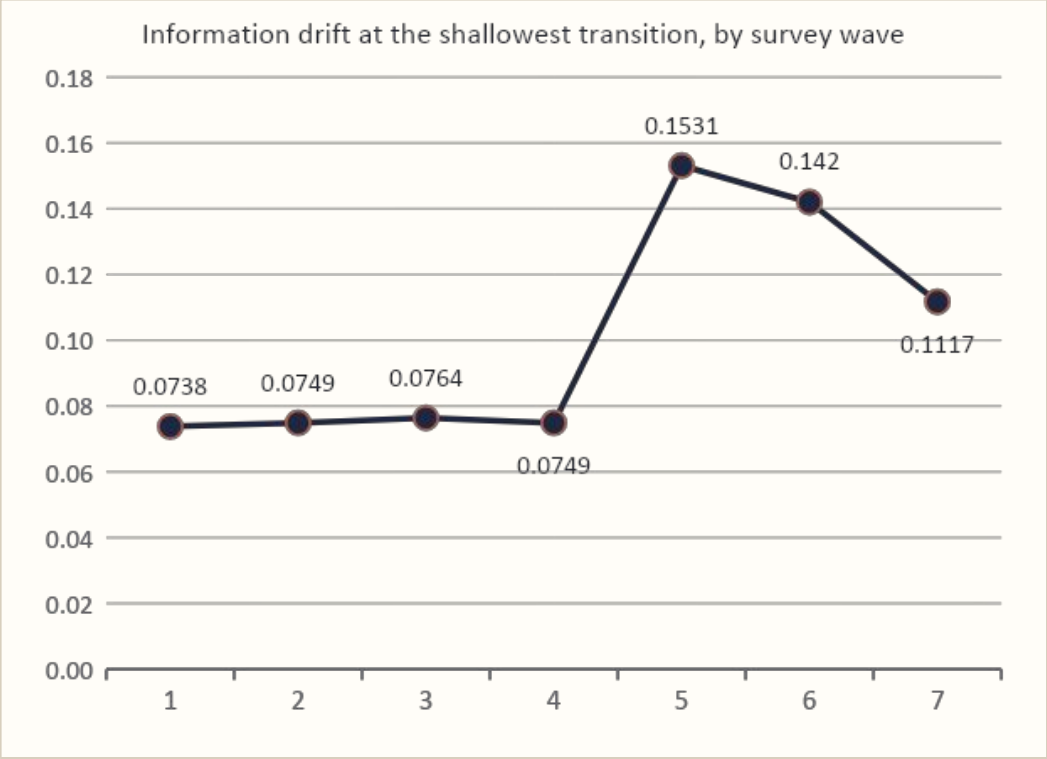


rows: structural depth d = 0, …, D−1 columns: the three probes

A task-free, multiscale fingerprint: repeated realizations of the same organization approach the same matrix, persistent displacement signals reorganization, and depth localizes where it occurs.

POSITIVE CONTROL

GSS: recovering a known data shift



0.0738–0.0764

2012–2018 · four waves, one design

Re-realization: nearly the same value

0.1531

2021 · primarily web-based

Shift ≈ 30× the 2012–2018 spread

0.1420 → 0.1117

2022, 2024 · mixed web/face-to-face

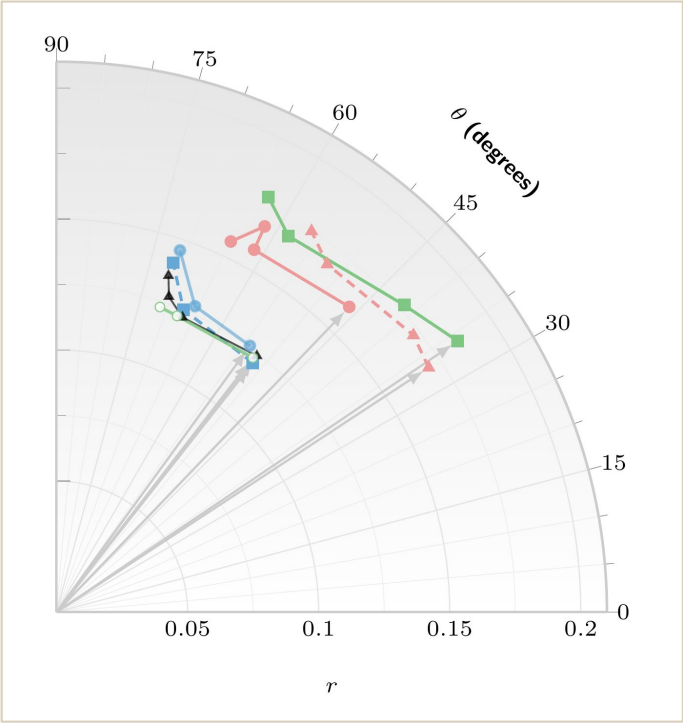
Partial return toward the earlier regime

Conditionals estimated with our LSM implementation, one instantiation of the conditional-learning objective rather than a requirement of the spectrum.

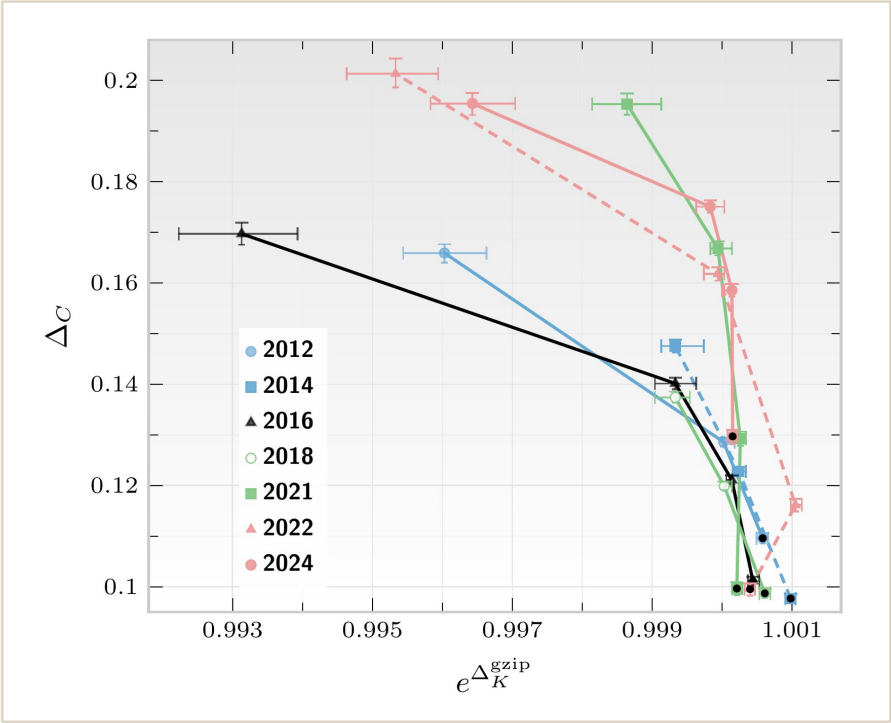
Repeated waves under one design give nearly the same fingerprint; the documented design change displaces it, without training to discriminate years or modes.

POSITIVE CONTROL

Two projections of the GSS spectrum



a Polar rendering of the raw (information drift, structural drift) plane. A tight 2012–2018 regime, a pronounced 2021 displacement, and partial return in 2022 and 2024.



b gzip/NCD information displacement against the Dirichlet-posterior mean of structural drift; the horizontal axis shows exp(Kolmogorov drift).

MULTISCALE COMPARISON

Spectra as fingerprints: 2018 vs 2024

$M_{2018} \approx$

	Δ_C	Δ_K^{pseudo}	Δ_K^{gzip}
$d = 0$	.0971	.0749	.00061
$d = 1$	.1130	.0460	.00004
$d = 2$	.1164	.0395	−.00090

$M_{2024} \approx$

	Δ_C	Δ_K^{pseudo}	Δ_K^{gzip}
$d = 0$	.1165	.1117	.00015
$d = 1$	.1383	.0754	−.00008
$d = 2$	.1472	.0794	−.00035

First three transitions; cell shading scales with value within each of the first two columns.

Reading the fingerprints

- Shallowest transition: information drift rises from .0749 (2018) to .1117 (2024).
- 2024 exceeds 2018 in both structural and information drift at all three depths.
- Structural drift grows with depth in both years; Kolmogorov drift stays below 0.001 in magnitude.
- Repeated observations can be compared through their full spectra rather than a single classifier or one global distance.

OUTLOOK

Interpretation and next steps

A system-level measurement object whose primary distinction is between re-realization and reorganization.

Its only learning requirement is a stable estimate of the target-wise conditionals. LSMs are one available realization; alternative learners can be substituted without changing the definition of the spectrum.

ROUTES FOR QUANTIFYING FINITE-SAMPLE VARIABILITY AND CALIBRATING EACH COORDINATE

Matched-size refit nulls

Separate finite-sample variability from real drift at matched sample sizes.

Independent peeling rules

Recover the hierarchy with alternative peeling rules to test robustness.

Planted-change simulations

Inject known changes to map the sensitivity of each coordinate.

Alternative conditional learners

Swap the conditional learner to test dependence on the algorithm.

Alternative compressors

Test how Kolmogorov drift depends on the choice of gzip.

Observation-design controls

Separate changes in how data are collected from changes in the system.

References

[1] J. G. Moreno-Torres et al., “A unifying view on dataset shift in classification,” Pattern Recognition 45(1):521–530, 2012.

[2] J. Gama et al., “A survey on concept drift adaptation,” ACM Computing Surveys 46(4), 2014.

[3] A. Gretton et al., “A kernel two-sample test,” JMLR 13:723–773, 2012.

[4] D. Lopez-Paz and M. Oquab, “Revisiting classifier two-sample tests,” ICLR, 2017.

[5] N. Sizemore et al., “A digital twin of the infant microbiome to predict neurodevelopmental deficits,” Science Advances 10(15):eadj0400, 2024.

[6] K. Wu, F. Li, and I. Chattopadhyay, “Emergenet: A digital twin of influenza A evolution for vaccine strain forecasting and emergence risk assessment,” Military Medicine, usag255, 2026.

[7] NORC at the University of Chicago, “General Social Survey 2021 codebook,” 2021.

[8] NORC, “General Social Survey 2022 codebook,” 2023.

[9] NORC, “General Social Survey 2024 codebook,” release 2, 2025.

[10] NORC, “Sequential data collection designs in the General Social Survey,” GSS Methodological Report 140, 2025.

[11] M. Li et al., “The similarity metric,” IEEE Trans. Information Theory 50(12):3250–3264, 2004.

Acknowledgment. This material is based upon work supported by the Defense Advanced Research Projects Agency (DARPA) under Agreement No. HR0011-26-3-E016. The views, opinions, and/or findings expressed are those of the authors and should not be interpreted as representing the official views or policies of the Department of Defense or the U.S. Government.

Correspondence: ishanu_ch@uky.edu